

DATA ANALYTICS | PORTFOLIO
PROJECT

Telco Customer Churn Analysis

Exploratory Data Analysis using Python, Pandas, Seaborn & Matplotlib

Data Analyst | Data Science

Python

Pandas

NumPy

Matplotlib

7,043

Total Customers

26.5%

Churn Rate

73.5%

Retention Rate

21

Features

Project Overview

Business Problem · Dataset · Methodology

7,043

Total Customers

26.5%

Churn Rate

73.5%

Retention Rate

21

Features

⚠ BUSINESS PROBLEM

Telecom companies lose significant revenue due to customer churn. Acquiring a new customer costs 5–7x more than retaining an existing one. This analysis identifies key patterns and risk factors driving churn.

🎯 PROJECT GOAL

Perform comprehensive EDA to uncover patterns in customer behavior, identify key churn drivers, and provide actionable business recommendations to improve retention strategy.

1

Data
Collection

2

Data
Cleaning

3

EDA &
Analysis

4

Key
Insights

5

Business
Recs

Data Understanding & Cleaning

Dataset Structure · Quality Assessment · Preprocessing

7,043

Rows

21

Columns

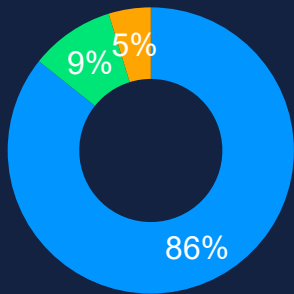
3

Numeric

18

Categorical

Data Type Distribution



■ Categorical (18) ■ Integer (2) ■ Float (1)

✓ TotalCharges → Numeric

`pd.to_numeric()` with `errors='coerce'` to handle whitespace values

✓ Missing Values Check

`df.isnull().sum()` revealed 11 NaN values in TotalCharges after conversion

✓ Drop Null Rows

`df.dropna(inplace=True)` → 7,032 clean records retained

✓ Target Encoding

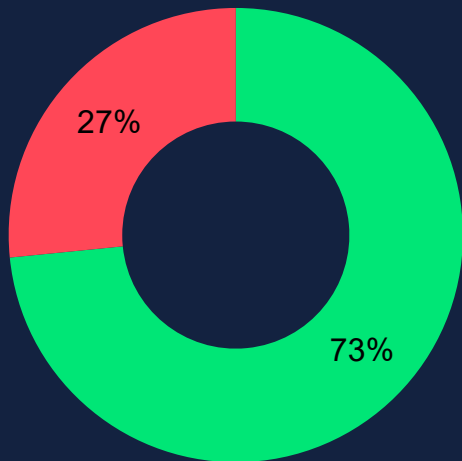
Churn: Yes → 1, No → 0 using `.map()` for numeric analysis

✓ **RESULT: Zero null values after cleaning | Final dataset: 7,032 rows × 21 columns**

Churn Distribution Analysis

Target Variable · Class Imbalance · Business Impact

Customer Churn Distribution



■ Retained (73.5%) ■ Churned (26.5%)

5,163

Retained Customers

1,869

Churned Customers

73.5%

Retention Rate

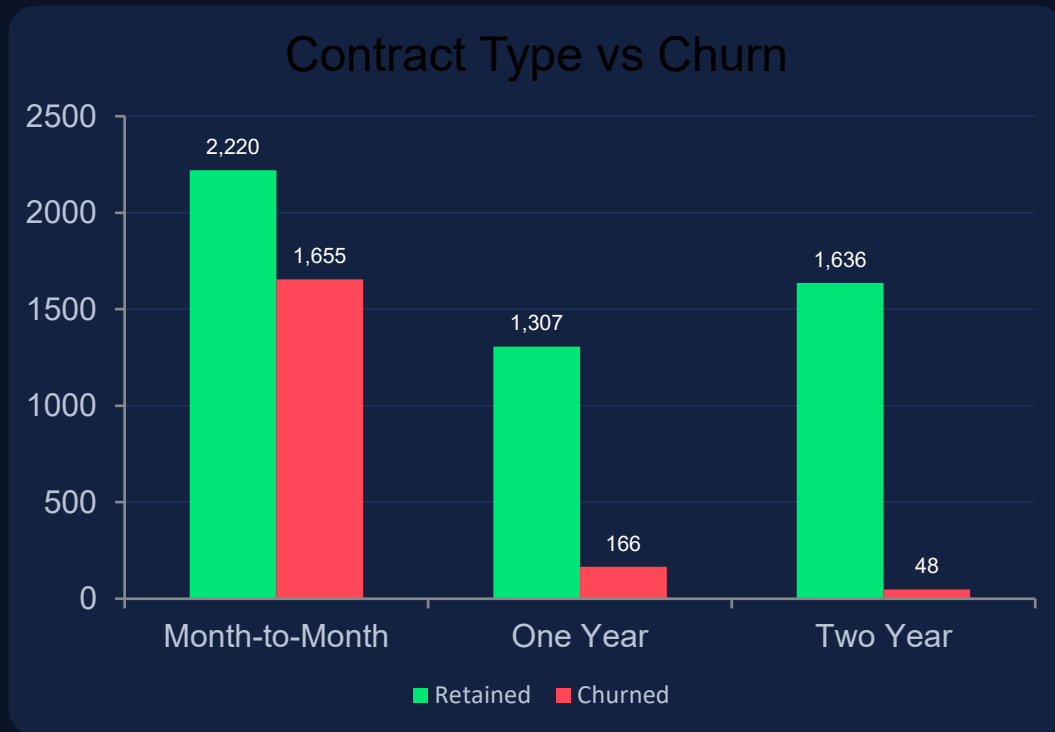
26.5%

Churn Rate

⚡ **IMBALANCE ALERT:** Dataset has ~3:1 ratio (retained:churned). Requires stratified sampling for ML models. 1 in 4 customers is leaving — immediate action needed!

Contract Type vs Churn KEY DRIVER

Month-to-Month · One Year · Two Year



Month-to-Month

~43% Churn

HIGH RISK

Longer contracts → Lower churn

One Year

~11% Churn

LOW RISK

Longer contracts → Lower churn

Two Year

~3% Churn

VERY SAFE

Longer contracts → Lower churn

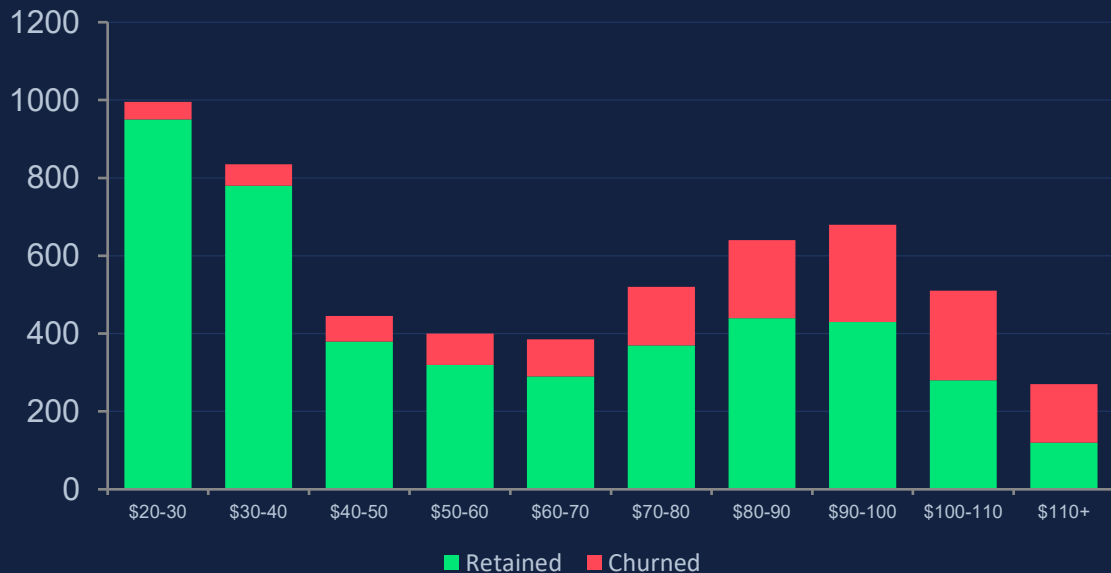


RECOMMENDATION: Convert monthly users to annual/two-year plans using loyalty discounts, free months, and exclusive perks — could reduce churn by up to 40%

Monthly Charges vs Churn

Price Sensitivity · Revenue Impact · Risk Zones

Monthly Charges Distribution by Churn



\$20 – \$40

~5% Churn Rate

Strong retention zone

\$60 – \$80

~28% Churn Rate

Moderate churn observed

\$80 – \$110

~45% Churn Rate

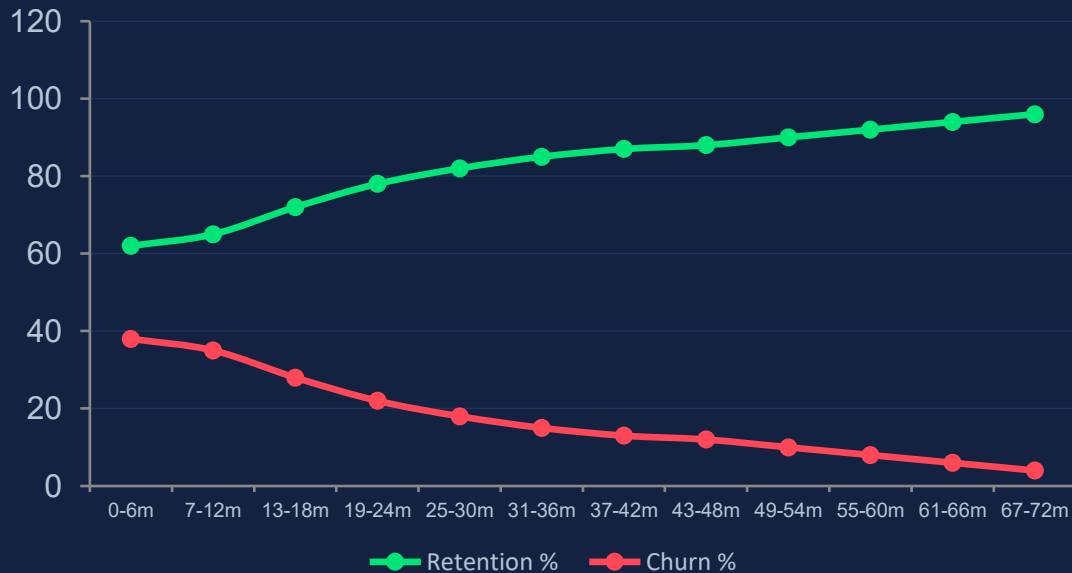
Peak churn zone

💡 **RECOMMENDATION:** Improve premium customer experience (\$80+). Offer loyalty discounts, value-added services, and better pricing perception to reduce churn in high-revenue segment.

Tenure vs Churn

Customer Lifecycle · Loyalty Curve · Critical Risk Period

Churn Rate by Tenure (months)



New Customer · 0–6 Months

CRITICAL

Highest churn — make-or-break period. Focus on onboarding.

Growth Phase · 7–24 Months

MODERATE

Churn declining. Engagement campaigns are effective.

Established · 25–48 Months

LOW

Stable customers. Loyalty rewards maintain retention.

Loyal Customer · 49–72 Months

VERY LOW



First 3–6 months are make-or-break. Invest heavily in onboarding, customer support, and early engagement to convert new users into long-term loyal customers.

Payment Method vs Churn

Electronic Check · Auto-Pay · Risk Indicators

Payment Method vs Churn



Electronic Check

~45% Churn

HIGHEST RISK

Mailed Check

~19% Churn

MODERATE

Auto-Pay (Bank/Card)

~16% Churn

SAFEST

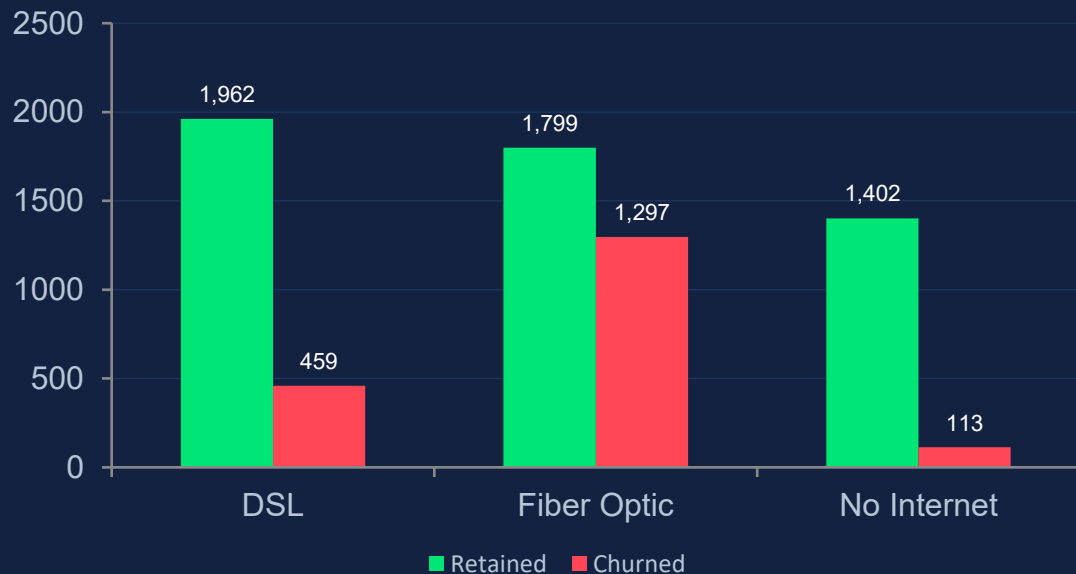


RECOMMENDATION: Encourage auto-payment through discounts, cashback, or fee waivers. Auto-pay customers show 3x better retention than electronic check users.

Internet Service vs Churn

Fiber Optic · DSL · Service Risk Segmentation

Internet Service Type vs Churn



Fiber Optic

41.9% Churn

High-value, high-risk. Likely cost/quality issues.

DSL

18.9% Churn

More stable segment. Cost-effective plans.

No Internet

7.4% Churn

Basic users — very low churn risk.

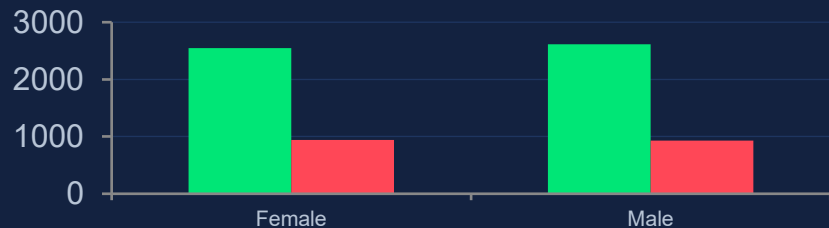


RECOMMENDATION: Fiber optic users are high-revenue but highest churn. Improve service reliability, reduce pricing relative to value, and offer dedicated fiber support.

Demographic Analysis

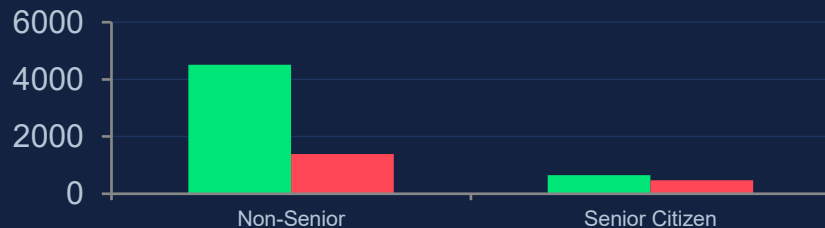
Gender · Senior Citizens · Partner · Dependents

Gender vs Churn



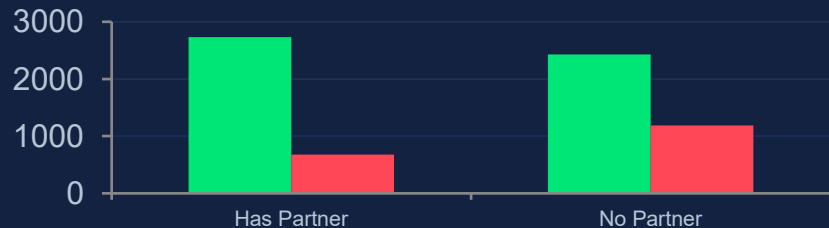
→ Gender has minimal impact on churn

Senior Citizen vs Churn



→ Seniors churn at ~42% — high-risk group

Partner vs Churn



→ No partner → higher churn risk

Dependents vs Churn

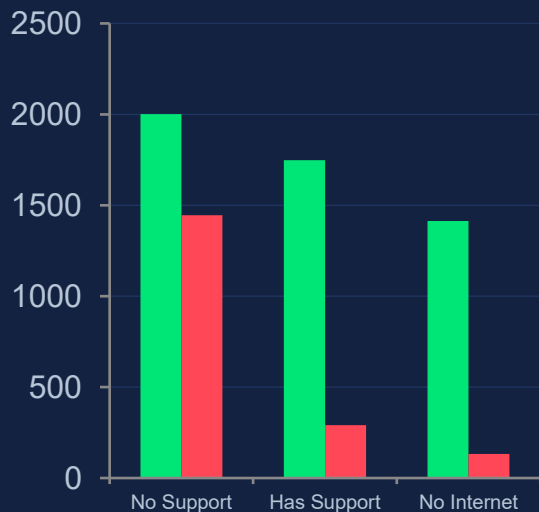


→ Customers with dependents are more loyal

Support Services vs Churn

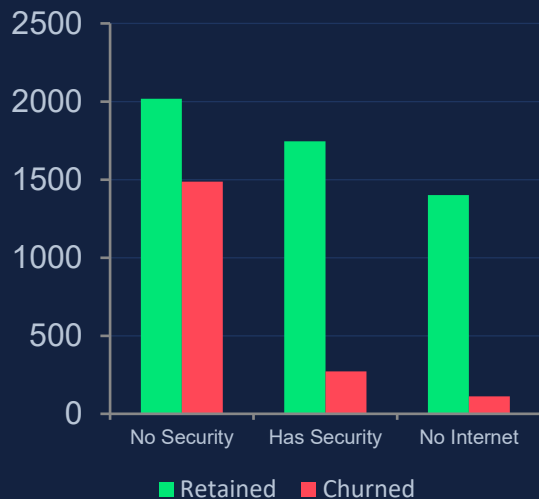
Tech Support · Online Security · Multiple Lines

Tech Support vs Churn



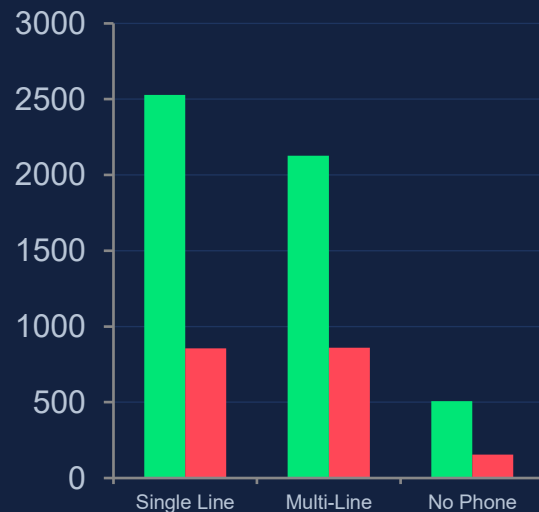
→ Tech support reduces churn by ~70%

Online Security vs Churn



→ Security adds confidence → strong retention

Multiple Lines vs Churn



→ Bundled lines create mild stickiness

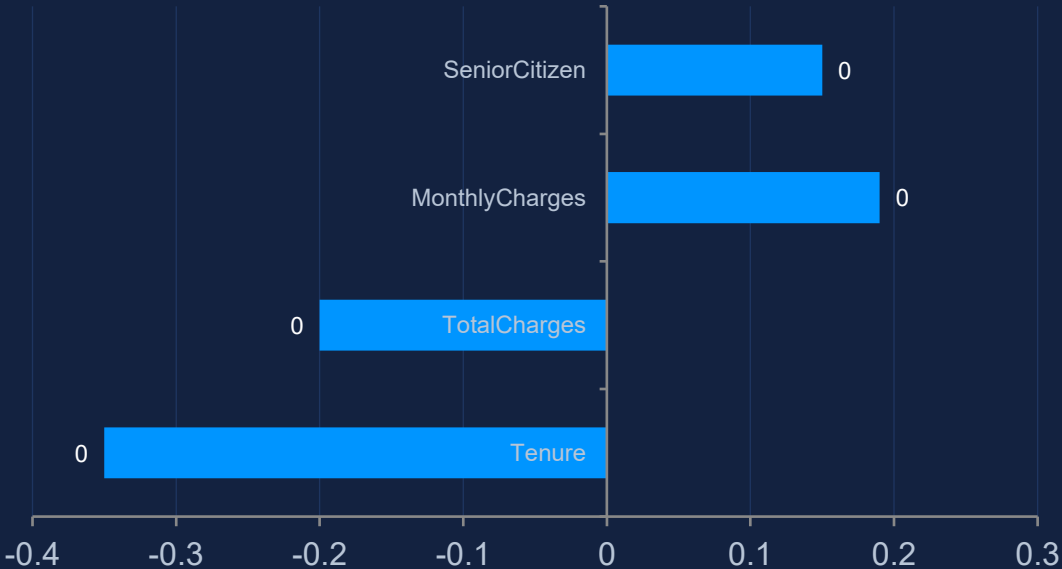


RECOMMENDATION: Bundle tech support + online security as standard add-ons. Free 3-month trials show significantly higher retention in customers who adopt bundled services.

Correlation Analysis

Feature Relationships · Churn Drivers · Key Correlations

Feature Correlation with Churn



Tenure ↔ Churn

-0.35

Strongest driver — longer stay = less churn

MonthlyCharges ↔ Churn

+0.19

Higher charges increase churn risk

TotalCharges ↔ Churn

-0.20

High total spenders are more loyal

SeniorCitizen ↔ Churn

+0.15

Slight but notable senior churn impact

Tenure ↔ TotalCharges

+0.83



KEY INSIGHT: Tenure is the most important predictor of churn ($r = -0.35$). Retaining customers through early stages dramatically reduces churn probability over time.

Key Business Insights

Executive Summary · Top Findings · Risk Factors

01 Month-to-Month = Highest Risk

~43% churn vs 3% for 2-year contracts. Contract type is the #1 churn driver.

02 New Customers Churn Most

First 3–6 months are critical. High acquisition without strong onboarding leads to early dropout.

03 High Charges = High Churn

Customers paying \$70–\$110/month show peak churn rates. Price sensitivity is significant.

04 Fiber Optic Users at Risk

41.9% churn — highest among internet service types. High cost + service issues drive attrition.

05 Auto-Pay Reduces Churn

Auto-payment users show 3x better retention than electronic check users.

06 Support Services Retain

Tech support and online security each reduce churn by ~70% within their segments.

Business Recommendations

Strategic Actions · ROI Impact · Action Roadmap

PRIORITY 1

Convert to Annual Plans

Offer 2-month free trial on annual/two-year contracts

📄 Reduce churn by up to 40%

PRIORITY 2

Strengthen Onboarding

Dedicated support for first 6 months — calls, tutorials, check-ins

📄 Cut early churn by 30%

PRIORITY 3

Promote Auto-Payment

Incentivize bank transfer/credit card auto-pay with discounts

📄 3x retention vs e-check

PRIORITY 4

Bundle Support Services

Include tech support + online security as standard in premium plans

📄 70% lower churn with support

PRIORITY 5

Fiber Optic Strategy

Improve QoS, address pricing concerns, dedicated fiber support team

📄 Reduce 41.9% churn segment

PRIORITY 6

Premium Customer Retention

Loyalty rewards + value-added services for \$80+/month customers

📄 Protect highest revenue segment

Thank You

Telco Customer Churn Analysis — EDA Complete

Key Takeaways

- Contract type is the #1 churn driver — monthly contracts risk 43% loss
- First 3–6 months are critical — strong onboarding is essential
- Auto-pay + bundled support services dramatically improve retention
- Fiber optic and high-charge segments need urgent strategy focus
- EDA reveals 6 high-impact intervention points for churn reduction

Future Scope

1. ML Churn Prediction Model
2. Customer Segmentation (K-Means)
3. Churn Prediction Dashboard
4. Real-time Risk Scoring API
5. A/B Test Retention Strategies

Tools Used:

Python · Pandas · NumPy · Matplotlib · Seaborn · Google Colab · PowerPoint